

tf.roll Stay organized with collections

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https://www.tensorflow.org/api_docs/python/tf/roll

Rolls the elements of a tensor along an axis.

Function Signatures

```
tf.roll( input, shift, axis, name=None )
```

Code Examples

```
tf.roll(  
input, shift, axis, name=None  
)
```

```
tf.roll(  
input, shift, axis, name=None  
)
```

```
# 't' is [0, 1, 2, 3, 4]  
roll(t, shift=2, axis=0) ==> [3, 4, 0, 1, 2]  
# shifting along multiple dimensions  
# 't' is [[0, 1, 2, 3, 4], [5, 6, 7, 8, 9]]  
roll(t, shift=[1, -2], axis=[0, 1]) ==> [[7, 8, 9, 5, 6], [2, 3, 4,  
0, 1]]  
# shifting along the same axis multiple times  
# 't' is [[0, 1, 2, 3, 4], [5, 6, 7, 8, 9]]  
roll(t, shift=[2, -3], axis=[1, 1]) ==> [[1, 2, 3, 4, 0], [6, 7, 8,  
9, 5]]
```

```
# 't' is [0, 1, 2, 3, 4]  
roll(t, shift=2, axis=0) ==> [3, 4, 0, 1, 2]  
# shifting along multiple dimensions  
# 't' is [[0, 1, 2, 3, 4], [5, 6, 7, 8, 9]]  
roll(t, shift=[1, -2], axis=[0, 1]) ==> [[7, 8, 9, 5, 6], [2, 3, 4,  
0, 1]]  
# shifting along the same axis multiple times  
# 't' is [[0, 1, 2, 3, 4], [5, 6, 7, 8, 9]]  
roll(t, shift=[2, -3], axis=[1, 1]) ==> [[1, 2, 3, 4, 0], [6, 7, 8,  
9, 5]]
```

Documentation

Rolls the elements of a tensor along an axis.

View aliases

Compat aliases for migration

See

Migration guide for
more details.

`tf.compat.v1.manip.roll`, `tf.compat.v1.roll`

Used in the notebooks

- DeepDream

The elements are shifted positively (towards larger indices) by the offset of shift along the dimension of axis. Negative shift values will shift elements in the opposite direction. Elements that roll passed the last position will wrap around to the first and vice versa. Multiple shifts along multiple axes may be specified.

For example:

`## Returns`